

MATH 1564 — Notes 09/17

Theorem. The usual algorithm for solving $Ax = 0$ and writing the solution in parametric vector form gives a basis for $\text{Nul } A$.

Theorem. The pivot columns of A form a basis for $\text{Col } A$.

Example. Let $A = [a_1 \ a_2 \ a_3 \ a_4]$, where

$$A = \begin{bmatrix} 1 & 1 & -1 & 1 \\ 2 & 3 & -1 & 0 \\ 2 & 3 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & -1 & 1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim R = \begin{bmatrix} 1 & 0 & -2 & 3 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The pivot columns are columns 1 and 2, so $\{a_1, a_2\}$ is a basis for $\text{Col } A$.

For $Rx = 0$, the free variables are x_3 and x_4 :

$$r_3 = -2r_1 + r_2, \quad r_4 = 3r_1 - 2r_2.$$

Thus the dependence relations among the columns of A are

$$2a_1 - a_2 + a_3 = 0, \quad -3a_1 + 2a_2 + a_4 = 0.$$

Equivalently,

$$\text{Nul } A = \text{span} \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

In particular, a_3 and a_4 are in $\text{span}\{a_1, a_2\}$.

Examples of bases

Example. What is a basis for $\{0\}$, the zero subspace of $\mathbb{R}^n$?

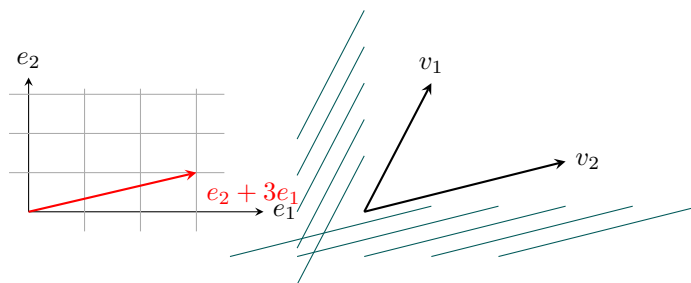
The empty set is a basis: it spans $\{0\}$, and it has no dependent vectors. The set $\{0\}$ itself is not a basis because the zero vector is always dependent.

Example. If S is a nonzero subspace of $\mathbb{R}^n$, how many bases does S have?

Infinitely many. If $\{v_1, \dots, v_k\}$ is a basis, then so are, for example,

$$\{2v_1, \dots, 2v_k\}, \quad \{3v_1, \dots, 3v_k\}, \quad \dots$$

Basis as a grid. A basis gives coordinate directions. In the standard basis of $\mathbb{R}^2$, e_1 and e_2 determine a rectangular grid; for example, $e_2 + 3e_1$ is located by moving 3 units in the e_1 direction and 1 unit in the e_2 direction. A nonstandard basis $\{v_1, v_2\}$ produces a skew grid, but it still gives unique coordinates.



Coordinates relative to a basis

Definition. Let $\beta = \{b_1, \dots, b_k\}$ be a basis for a subspace S . For $x \in S$, its coordinates relative to β are the scalars $c_1, \dots, c_k$ such that

$$x = c_1 b_1 + \dots + c_k b_k.$$

We write

$$[x]_\beta = \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix}.$$

Example. Let

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Then $\{v_1, v_2\}$ is a basis of $\mathbb{R}^2$: indeed,

$$e_1 = \frac{v_1 + v_2}{2}, \quad e_2 = \frac{v_1 - v_2}{2}.$$

Thus the vectors span $\mathbb{R}^2$, and their independence is clear.

Find the coordinates of $e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ in $\beta = \{v_1, v_2\}$. If

$$[e_1]_\beta = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix},$$

then

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.$$

Theorem. x and β determine $[x]_\beta$.

Proof. Suppose $\beta = \{b_1, \dots, b_k\}$ and

$$x = c_1 b_1 + \dots + c_k b_k = d_1 b_1 + \dots + d_k b_k.$$

Then

$$\begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix} = \begin{bmatrix} d_1 \\ \vdots \\ d_k \end{bmatrix},$$

because β is linearly independent. Hence the coordinate vector is unique.

Dimension

Definition. The dimension of a subspace $S \subseteq \mathbb{R}^n$ is the number of vectors in a basis of S .

Examples.

- $\dim(\mathbb{R}^n) = n$, since $\{e_1, \dots, e_n\}$ is a basis.
- $\dim\{0\} = 0$.
- $\dim(\text{Col } A)$ is the number of pivots of A .
- $\dim(\text{Nul } A)$ is the number of free variables, equivalently the number of nonpivot columns.

For

$$S = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n : x_1 + \dots + x_n = 0 \right\},$$

we have $S = \text{Nul } A$, where $A = \begin{bmatrix} 1 & \dots & 1 \end{bmatrix}$. Since there is one pivot and $n - 1$ free variables,

$$\dim S = n - 1.$$

Theorem. Let S be a subspace of $\mathbb{R}^n$. Any two bases of S have the same number of vectors.

Proof. Let $\{a_1, \dots, a_k\}$ and $\{b_1, \dots, b_\ell\}$ be bases of S . Assume, for a contradiction, that $k > \ell$. Since the b_i span S , write

$$a_i = c_{i1}b_1 + \dots + c_{i\ell}b_\ell = Bc_i,$$

where

$$B = [b_1 \ \dots \ b_\ell], \quad C = [c_1 \ \dots \ c_k].$$

Then

$$A = [a_1 \ \dots \ a_k] = BC.$$

The columns of A are independent because they form a basis, so the columns of C must be independent. Thus C , which has only ℓ rows, would need k pivot columns. This is impossible when $\ell < k$. Therefore $k \leq \ell$; reversing the roles of the two bases gives $\ell \leq k$, so $k = \ell$.

Example. If A is a 5×7 matrix, then

$$\dim(\text{Col } A) \leq 5, \quad \dim(\text{Nul } A) \geq 2.$$

Definition. The rank of A is the number of pivots of A .

If $T(x) = Ax$, then the image of T is $\text{Col } A$, which is a subspace, and

$$\dim(\text{image } T) = \text{rank}(A).$$

Rank Theorem. If $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map, then

$$\dim(\text{image } T) + \dim(\ker T) = n.$$

Proof. Let $T(x) = Ax$. Then

$$\text{image } T = \text{Col } A, \quad \ker T = \text{Nul } A.$$

The dimension of $\text{Col } A$ is the number of pivot columns, and the dimension of $\text{Nul } A$ is the number of nonpivot columns. Every column is one or the other, so their dimensions add to n .

Example. Let

$$T(x_1, x_2, x_3) = 2x_1 - x_2 + x_3.$$

This is linear because $T(x) = Ax$, where

$$A = \begin{bmatrix} 2 & -1 & 1 \end{bmatrix}.$$

Then

$$\dim(\text{image } T) = \dim(\text{Col } A) = 1, \quad \dim(\ker T) = \dim(\text{Nul } A) = 2.$$

Recall: a basis of a subspace S is a set of independent vectors that spans S .

Theorem. Let $v_1, \dots, v_k \in S$, where $\dim S = k$. The following are equivalent:

1. $\{v_1, \dots, v_k\}$ is a basis of S .
2. $S = \text{span}\{v_1, \dots, v_k\}$.
3. $\{v_1, \dots, v_k\}$ is linearly independent.

By definition, (1) implies both (2) and (3), and (2) together with (3) implies (1).

For (2) $\Rightarrow$ (3), the pivot columns of $[v_1 \cdots v_k]$ form a basis of $\text{Col}[v_1 \cdots v_k] = S$. Since $\dim S = k$, there are k pivots, so every column is pivotal.

For (3) $\Rightarrow$ (2), if $\text{span}\{v_1, \dots, v_k\} \neq S$, choose $v_{k+1} \in S$ outside that span. Then $\{v_1, \dots, v_k, v_{k+1}\}$ is independent. Repeating would create an independent set in $\mathbb{R}^n$ with more than n vectors, which is impossible.

Let $S_i = \text{span}\{v_1, \dots, v_i\}$, with $\{v_1, \dots, v_i\}$ independent. Then

$$S_k \subsetneq S_{k+1} \subsetneq S_{k+2} \subsetneq \dots \subseteq S \subseteq \mathbb{R}^n.$$

Lemma. $\mathbb{R}^n$ contains at most n linearly independent vectors.

Proof. If $\{v_1, \dots, v_k\} \subseteq \mathbb{R}^n$ is independent, then the $n \times k$ matrix $[v_1 \ \dots \ v_k]$ has k pivots. Hence $k \leq n$.

Eventually, the process above must stop. At that point $S_i = S$, so $\{v_1, \dots, v_i\}$ is a basis of S . This proves the equivalence theorem.

Theorem. Every nonzero subspace of $\mathbb{R}^n$ has a basis.

Proof. Pick any nonzero $v_1 \in S$ and let $S_1 = \text{span}\{v_1\}$. If $S = S_1$, then $\{v_1\}$ is a basis. Otherwise, choose $v_2 \in S \setminus S_1$; then $\{v_1, v_2\}$ is independent. Continue: if

$$S_i = \text{span}\{v_1, \dots, v_i\} \neq S,$$

choose $v_{i+1} \in S \setminus S_i$. The lemma guarantees that this cannot continue past n independent vectors. Therefore it stops with $S_i = S$, giving a basis.